Symbolic Music Data Analysis Engine

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May 2024

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Introduction

Original Goal: Identify the melodies of a polyphonic music piece with heuristics.

Current Approaches:

- Skyline: Find Highest Pitch (G. Ozcan, C. Isikhan, A. Alpkocak, 2005)

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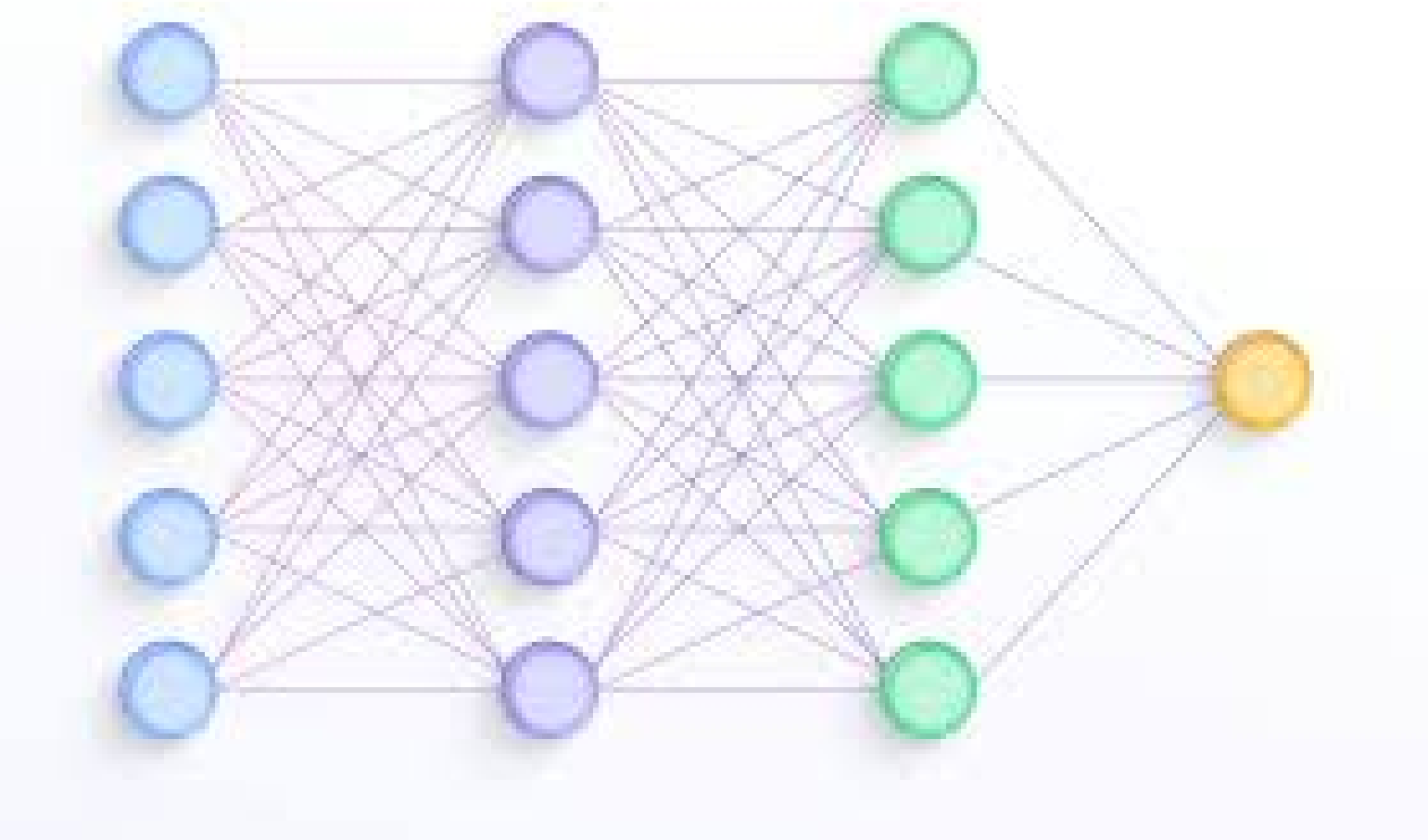
- Skyline: Find Highest Pitch (G. Ozcan, C. Isikhan, A. Alpkocak, 2005)
- Deep Learning (Y. Chou, I. Chen, C. Chang, J. Ching and Y. Yang, 2021)

Skyline



Figure 1. Illustration of our baselines. a) Input example, b) Output of skyline algorithm, c) Output of skyline variation algorithm.

Neural Network



Evolution

After further analysis of this engine, we realized that a generic heuristics-based approach is capable of many more tasks than simple melody identification.

As a result, one other use we identified, with a separate set of heuristics, is to identify, measure by measure, which measures deserve the attention of the conductor.

We believe that many more uses and many more heuristics can be encoded, as such this tool could be a much more generalized tool than currently shown.

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Heuristics Identified (Melody Extraction)

Pre-processing

Remove all chords and all percussion first.

- Loudest Dynamic
- Highest Pitch
- Significant Pitch Delta
- Short Rhythm
- Articulation Marks
- Continuity

With these heuristics we are able to assign a score to each of the measures.

Example



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Overview

The algorithm iterates through the symbolic music measure by measure. For each measure, it looks at each instrument at that measure. Suppose we have at measure i, instruments $[m_1, m_2, ...m_k]$.

For each instrument m at measure i, we extract its features and evaluate based on the weights we provide, and a "likelihood value" is obtained that shows how likely it is to be melody.

Then, for measure i, we sort the "likelihood value" to determine which instrument(s) is most likely to be melody and for that particular measure, we make that instrument to be melody. Repeat this for every measure.

Psuedocode

```
k := number of parts we keep
K_i := the symbolic music (i.e. the measure) we intend to keep
W := weights for each heuristic
S_im := vector of extracted heuristic values at measure i for
S_i := vector storing the weight, dot product each S_im, that
R := new score used to store the extracted music
---------------

for each measure i in the piece:
    S_i = []
    for each instrument m in measure:
        S_im = extract_heuristic()
        S_i.append(W @ S_im) # S_i = [S_i, W@S_im]
    K_i = sorted(S_i)[1:k]
    R.append(K_i)
return R
```

Feature Extraction

We want to normalize each of the heuristics to a 0-1 value.

Recall that articulation, rhythm shortness and continuity are already normalized as they are True/False values.

Thus, we can either normalize with a discrete continuous function.

$$f(pitch) = \begin{cases} 1 & \text{if pitch} > \text{mode or pitch} = \max \\ 0 & \text{otherwise} \end{cases}$$

Or

$$f(p\delta_i) = \frac{p\delta_i}{\text{argmax}_{instruments}(p\delta)}$$

Training

We use Multiplicative Weight Update method, a method widely used in Machine Learning and Game Theory, to train our weights.

We update the weight of each wrong heuristic w_i at each iteration to be

$$w_i^{t+1} = w_i^t(1 - \lambda)$$

where λ is a fixed constant. We then normalize all the weights such that they sum to 1.

This gives us a guarantee in expectation polynomially bound by m^* minimum error and n number of heuristics. (S. Arora, E. Hazan, S. Kale, 2012)

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Results

We tested this algorithm on a section Dvorak's New World Symphony Movement IV. The accuracy score for our method is filled in from this test.

Heuristic	Skyline	CNN
88.9%	78.7%	91.7%

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Improvements

We find that while such an expert system is generally better than the single heuristic Skyline, it has the following weaknesses.

- Missing Pickups (Fixed in Demo.)
- Long Ending Notes
- Counter-melody
- Absolute Volume

Additionally, some finer grained control could be introduced in place of coarse heuristics such as pitch delta. Some additional heuristics such as sectional similarity can also be employed.

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Approach

For this task, we note that for a conductor, as opposed to the fixed number of melodies expected in a piece, a measure deserves attention from the conductor *as soon as one particular feature is prominent*.

As such, we still employ the heuristics-based approach, however, as soon as one of the heuristics is triggered, we mark the measure for the conductor.

Heuristics

We first acknowledge that this list of potential heuristics is far from comprehensive. However, we attempt to identify measures that have

- Changing Time or Key Signature
- Difficult Ranges of Pitches
- Difficult Rhythm
- Accidentals

Annotation Demo

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Future Improvements

We note that the weaknesses previously mentioned is more prominent in the Annotation task.

We note that as an example, the problem of lack of fine grained note-by-note control significantly limits the effectiveness of some features that could be implemented, such as dissonance analysis.

Limitations

Furthermore, there may be further limitations with the heuristics approach, mainly that many important but more complex features are difficult to identify.

One example of such a feature would be the idea of a "call and response", or "duet". It is very difficult to define a set of criteria to identify this feature, yet this feature would be very useful for both melody detection and annotation.

We suggest that some of these heuristics could be identified with either a set of simpler heuristics or machine learning methods. It deserves more discussion on this topic, however.

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Acknowledgements

I'd like to graciously acknowledge the guidance from Dr. Chuck-jee Chau of the Chinese University of Hong Kong throughout this project, as I complete my exchange term in Hong Kong this Winter term.

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Y. Chou, I. Chen, C. Chang, J. Ching, Y. Yang, MidiBERT-Piano: Large-scale Pre-training for Symbolic Music Understanding, arXiv, 2021, 2107.05223

Music Used

Liz and the Blue Bird, Movement III and IV, Akito Matsuda, Arranged by ilovebees, https://musescore.com/ilovebees/liz-and-the-blue-bird

The Symphony No. 9 in E minor, "From the New World", Op. 95, B. 178, Movement IV, Antonín Dvořák, IMSLP